

# Chapter 1 Basics concepts of hydrological models: routing

Olivier Bonte

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To get more insight into the routing concepts explained in the theory, a computational guide is given here, leveraging both symbolic and numerical computing.

## 1 Reservoir cascade models

### 1.1 Cascade of linear reservoirs

#### 1.1.1 First order linear reservoir with constant input

A linear reservoir with storage  $S$  [m<sup>3</sup>] and a constant input  $I$  [m<sup>3</sup>/s] is characterised by

$$S = KQ \tag{1}$$

with  $K$  [s] the residence time and  $Q$  [m<sup>3</sup>/s] the outflow. The mass balance equation is given by

$$\frac{dS}{dt} = K \frac{dQ}{dt} = I - Q \tag{2}$$

As this is a linear ordinary differential equation, it can be solved analytically using Laplace transforms (see your course in the 2nd bachelor year on [Differential Equations](#)). Remember that although the definition of a Laplace transform might seem complex

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}dt,$$

in practice you only need to apply a limited set of transforms to solve linear ODEs, which are summarised in the table below.

|          |                              |
|----------|------------------------------|
| $f(t)$   | $F(s) = \mathcal{L}\{f(t)\}$ |
| $H(t)$   | $\frac{1}{s}$                |
| $t^n$    | $\frac{n!}{s^{n+1}}$         |
| $e^{at}$ | $\frac{1}{s-a}$              |
| $f'(t)$  | $sF(s) - f(0)$               |

where  $H(t)$  is the Heaviside function, which is defined as  $H(t) = 0$  for  $t < 0$  and  $H(t) = 1$  for  $t \geq 0$ .  $n$  is a positive integer and  $a$  is a real number. Applying the Laplace transform to equation Equation 2 with  $Q(t=0) = Q_0 = S_0/K$  and  $q(s) = \mathcal{L}\{Q(t)\}$  gives:

$$K(sq(s) - Q_0) = \frac{I}{s} - q(s)$$

Isolating  $q(s)$  gives:

$$q(s) = \frac{I}{s(1+Ks)} + \frac{KQ_0}{1+Ks} \quad (3)$$

We now need to apply the inverse Laplace transform to get  $Q(t)$ . Trivially,  $\mathcal{L}^{-1}(q(s)) = Q(t)$ . For the term containing  $Q_0$ :

$$\mathcal{L}^{-1}\left(\frac{KQ_0}{1+Ks}\right) = \frac{K}{K}Q_0\mathcal{L}^{-1}\left(\frac{1}{1/K+s}\right) = Q_0e^{-t/K} \quad (4)$$

The term with  $I$  is a bit more convolved, but can be solved using partial fraction decomposition:

$$\frac{I}{s(1+Ks)} = \frac{A}{s} + \frac{B}{1+Ks}$$

To solve for  $A$ , multiply both sides by  $s$  and set  $s = 0$ :

$$\left.\frac{I}{1+Ks}\right|_{s=0} = A \iff A = I$$

To solve for  $B$ , multiply both sides by  $(1+Ks)$  and set  $s = -1/K$ :

$$\left.\frac{I}{s}\right|_{s=-1/K} = B \iff B = -IK$$

Now we can apply the inverse laplace transform on the partial fraction decomposition:

$$\mathcal{L}^{-1}\left(\frac{I}{s} - \frac{IK}{1+Ks}\right) = \mathcal{L}^{-1}\left(\frac{I}{s} - \frac{I}{s+1/K}\right) = I - Ie^{-t/K} = I(1 - e^{-t/K}) \quad (5)$$

Combining Equation 4 and Equation 5 gives the final solution for  $Q(t)$ :

$$Q(t) = I(1 - e^{-t/K}) + Q_0 e^{-t/K} \quad (6)$$

E.g. for  $Q_0 = 0.1 \text{ m}^3/\text{h}$  and  $K = 7 \text{ h}$ , we can plot the outflow  $Q(t)$  for different constant inflows  $I$ . In the examples that follow, we'll use  $\text{m}^3/\text{h}$  and  $\text{h}$  for the units instead of the standard SI-units for convenience. Define the analytical solutions as a function:

```
import matplotlib.pyplot as plt
import numpy as np

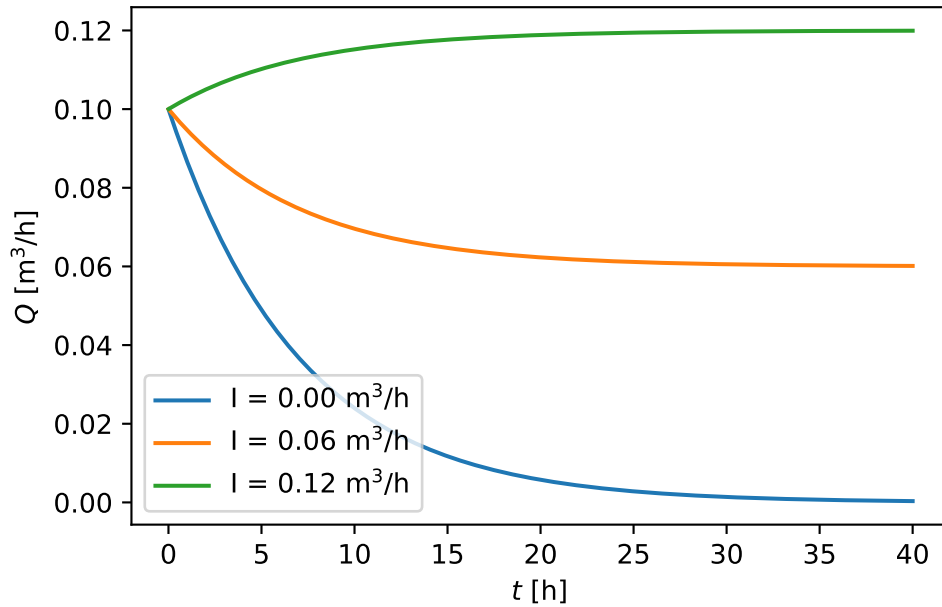
def Q_analytical_linear(t, I, K, Q0):
    return I * (1 - np.exp(-t / K)) + Q0 * np.exp(-t / K)
```

define the parameters and the three different inflows:

```
Q_0_num = 0.1 # m^3/h
K_num = 7 # h
t_array = np.linspace(0.01, 40, 200) # h
I_array = [0.0, 0.06, 0.12] # m^3/h
```

to then plot the results:

```
fig, ax = plt.subplots()
colors = plt.rcParams["axes.prop_cycle"].by_key()["color"]
for i, I_ in enumerate(I_array):
    Q_ = Q_analytical_linear(t_array, I_, K_num, Q_0_num)
    ax.plot(t_array, Q_, color=colors[i], label=rf"I = {I_:.2f} m$^3$/h")
ax.set_xlabel(r"$t$ [h]")
ax.set_ylabel(r"$Q$ [m$^3$/h]")
ax.legend()
```



Additionally, note that we can automate the algebraic routines above using [SymPy](#). First define all symbols:

```
import sympy as sp

K, I, Q0 = sp.symbols("K I Q0", positive=True)
t = sp.symbols("t", positive=True)
s = sp.symbols("s")
q, Q = sp.symbols("q, Q", cls=sp.Function)
```

and then define the ODE in the form  $K \frac{dQ}{dt} - (I - Q) = 0$ :

```
ode_linear = K * sp.diff(Q(t), t) - (I - Q(t))
ode_linear
```

$$-I + K \frac{d}{dt} Q(t) + Q(t)$$

Now take the Laplace transform of the ODE defining  $q(s) = \mathcal{L}\{Q(t)\}$ :

```
ode_linear_laplace = sp.laplace_transform(ode_linear, t, s, noconds=True)
ode_linear_laplace = sp.laplace_correspondence(ode_linear_laplace, {Q: q})
ode_linear_laplace
```

$$-\frac{I}{s} + K(sq(s) - Q(0)) + q(s)$$

We now insert the initial condition  $Q(0) = Q_0$ :

```
ode_linear_laplace_ic = sp.laplace_initial_conds(ode_linear_laplace, t, {Q:
    ↳ [Q0]})
ode_linear_laplace_ic
```

$$-\frac{I}{s} + K(-Q_0 + sq(s)) + q(s)$$

In the equation above, the only unknown is  $q(s)$ , which can be solved for:

```
q_s_solved = sp.solve(ode_linear_laplace_ic, q(s))[0]
q_s_solved
```

$$\frac{I + KQ_0s}{s(Ks + 1)}$$

Now apply the inverse Laplace transform to get  $Q(t)$ :

```
Q_solved = sp.simplify(sp.inverse_laplace_transform(q_s_solved, s, t))
Q_solved
```

$$\left(Ie^{\frac{t}{K}} - I + Q_0\right)e^{-\frac{t}{K}}$$

Which is identical to the analytical solution Equation 6 derived above.

### 1.1.2 Second order linear reservoir with constant input

We now get two reservoirs in series:

$$K^{(1)} \frac{dQ^{(1)}}{dt} = I - Q^{(1)}$$

$$K^{(2)} \frac{dQ^{(2)}}{dt} = Q^{(1)} - Q^{(2)}$$

The same SymPy approach can be extended to solve this coupled system. To simplify the solution, we'll assume:

- The second reservoir is initially empty, i.e.  $Q^{(2)}(0) = 0$ .
- The first reservoir has an initial outflow  $Q^{(1)}(0) = Q_0$  as it is assumed non-empty.
- The first and second reservoirs have the same residence time, i.e.  $K^{(1)} = K^{(2)} = K$ .

We define two unknown functions and their Laplace transforms:

```
q1, q2, Q1, Q2 = sp.symbols("q1 q2 Q1 Q2", cls=sp.Function)
```

Write both ODEs in the form  $(\dots) = 0$ :

```
ode1 = K * sp.diff(Q1(t), t) - (I - Q1(t))
ode2 = K * sp.diff(Q2(t), t) - (Q1(t) - Q2(t))
display(ode1, ode2)
```

$$-I + K \frac{d}{dt} Q_1(t) + Q_1(t)$$

$$K \frac{d}{dt} Q_2(t) - Q_1(t) + Q_2(t)$$

Take the Laplace transform using  $q_1(s) = \mathcal{L}\{Q_1(t)\}$  and  $q_2(s) = \mathcal{L}\{Q_2(t)\}$ :

```
ode1_laplace = sp.laplace_transform(ode1, t, s, noconds=True)
ode1_laplace = sp.laplace_correspondence(ode1_laplace, {Q1: q1})

ode2_laplace = sp.laplace_transform(ode2, t, s, noconds=True)
ode2_laplace = sp.laplace_correspondence(ode2_laplace, {Q1: q1, Q2: q2})
```

Insert the initial conditions:

```
ode1_laplace_ic = sp.laplace_initial_conds(ode1_laplace, t, {Q1: [Q0], Q2:
    ↪ [0]})
ode2_laplace_ic = sp.laplace_initial_conds(ode2_laplace, t, {Q1: [Q0], Q2:
    ↪ [0]})
display(ode1_laplace_ic, ode2_laplace_ic)
```

$$-\frac{I}{s} + K(-Q_0 + sq_1(s)) + q_1(s)$$

$$Ksq_2(s) - q_1(s) + q_2(s)$$

Solve the algebraic system for  $q_1(s)$  and  $q_2(s)$ :

```
q_s_solved_2 = sp.solve([ode1_laplace_ic, ode2_laplace_ic], [q1(s), q2(s)],
    ↪ dict=True)[0]
q2_s = q_s_solved_2[q2(s)]
q2_s
```

$$\frac{I + KQ_0s}{K^2s^3 + 2Ks^2 + s}$$

To get a solution closer to the notation of the theory, we'll first break up into partial fractions:

```
q2_s_partial = sp.apart(q2_s, s)
q2_s_partial
```

$$-\frac{IK}{Ks+1} + \frac{I}{s} - \frac{K(I-Q_0)}{(Ks+1)^2}$$

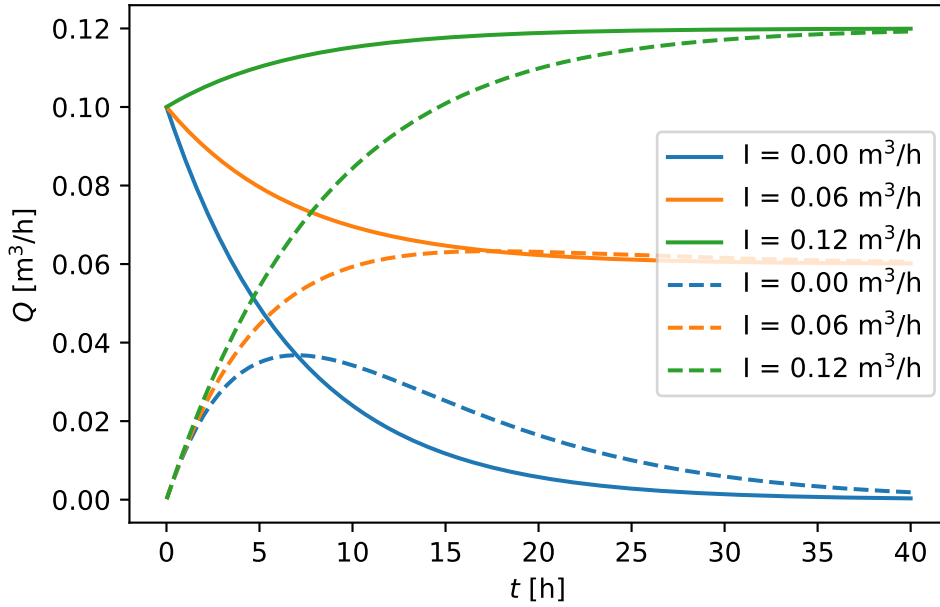
Finally, apply the inverse Laplace transform to obtain  $Q_2(t)$ .

```
Q2_solved = sp.inverse_laplace_transform(q2_s_partial, s, t)
Q2_solved
```

$$I - Ie^{-\frac{t}{K}} - \frac{t(I - Q_0)e^{-\frac{t}{K}}}{K}$$

Also  $Q_2(t)$  can now be visualised for different constant inflows  $I$  and the same parameters as before:

```
for i, I_ in enumerate(I_array):
    Q2_func = sp.lambdify(t, Q2_solved.subs({I: I_, K: K_num, Q0: Q_0_num}),
        ↪ "numpy")
    Q2_ = Q2_func(t_array)
    ax.plot(
        t_array, Q2_, label=rf"I = {I_:.2f} m$^3$/h", color=colors[i],
        ↪ linestyle="--"
    )
ax.set_xlabel(r"$t$ [h]")
ax.set_ylabel(r"$Q$ [m$^3$/h]")
ax.legend()
fig
```



In this figure, full lines are the output of the first reservoir and dashed lines are the output of the second reservoir. Note that the second reservoir has a delayed response compared to the first reservoir.

Given that the second reservoir is initially empty and receives no external input  $I$ ,  $Q_2$  is entirely defined by  $Q_1$  and  $K^{(1)} = K$ . So if you solve for  $Q_2$  (in the  $s$ -domain  $q_2(s)$ ) with an assumed known  $Q_1(t)$  function (in the  $s$ -domain  $q_1(s)$ ), you get:

```
q2_s_bis = sp.solve(ode2_laplace_ic, q2(s))[0]
q2_s_bis
```

$$\frac{q_1(s)}{Ks + 1}$$

This concept can be generalised in what is called a **transfer function**, which is defined as the ratio of the Laplace transforms of the output and input of a system. In this case, the transfer function  $H(s)$  is defined as:

$$H(s) = \frac{q_2(s)}{q_1(s)} = \frac{1}{1 + Ks} \quad (7)$$

### 1.1.3 $n$ -th order linear reservoir with constant input

The concept of 2 reservoirs can be generalised to  $n$  reservoirs in series, with the general form of the ODEs given by:

$$\frac{dS^{(i)}}{dt} = K \frac{dQ^{(i)}}{dt} = (Q^{(i-1)} - Q^{(i)}), \quad i = 1, 2, \dots, n$$

Again, we assume  $K^{(i)} = K$  for all reservoirs,  $Q^{(0)}(t) = I$  and  $Q^{(i)}(t=0) = 0 \quad \forall i > 1$ . The easiest way to solve this equation is to use the transfer function concept. The first reservoir is a special case, as it has an external input  $I$  and an initial outflow  $Q^{(1)}(0) = Q_0$ . In the  $s$ -domain,  $q^{(1)}(s)$  is given by Equation 3.  $\forall i > 1$ , the transfer function is given by a form analogous to Equation 7:

$$H^{(i)}(s) = \frac{q_i(s)}{q_{i-1}(s)} = \frac{1}{1 + Ks} \quad (8)$$

Consequently,  $q^{(n)}(s)$  can be expressed as:

$$q^{(n)}(s) = q^{(1)}(s) \prod_{i=2}^n H^{(i)}(s) = \frac{q^{(1)}(s)}{(1 + Ks)^{n-1}} \quad (9)$$

In SymPy, this can be expressed as (reusing the previously solved  $q^{(1)}(s)$  denoted as `q_s_solved`):

```
n = sp.symbols("n", positive=True, integer=True)
q_n_s = q_s_solved / ((1 + K * s) ** (n - 1))
q_n_s = sp.refine(q_n_s, sp.Q.gt(n, 1))
display(q_n_s)
```

$$\frac{(I + KQ_0s)(Ks + 1)^{1-n}}{s(Ks + 1)}$$

The inverse Laplace transform can be applied to get  $Q^{(n)}(t)$ :

```
Q_n_solved = sp.refine(sp.inverse_laplace_transform(q_n_s, s, t), sp.Q.gt(n,
↪ 1))
Q_n_solved
```

$$I\mathcal{L}_s^{-1} \left[ \frac{(Ks + 1)^{1-n}}{Ks^2 + s} \right] (t) + KQ_0 \frac{\partial}{\partial t} \mathcal{L}_s^{-1} \left[ \frac{(Ks + 1)^{1-n}}{Ks^2 + s} \right] (t)$$

Unfortunately, we run into a limitation of SymPy here, as it can't generalise the inverse Laplace transform for arbitrary  $n$  (try calling `sp.simplify` on `Q_n_solved`, it will crash). For specific values of  $n$ , we can still get the solutions

```

n_num_list = [5, 6, 7]
for n_num in n_num_list:
    print(f"n = {n_num}")
    Q_n_solved_specific = sp.simplify(Q_n_solved.subs(n, n_num))
    display(Q_n_solved_specific)

```

n = 5

$$I \left( 1 - e^{-\frac{t}{K}} - \frac{te^{-\frac{t}{K}}}{K} - \frac{t^2e^{-\frac{t}{K}}}{2K^2} - \frac{t^3e^{-\frac{t}{K}}}{6K^3} - \frac{t^4e^{-\frac{t}{K}}}{24K^4} \right) + \frac{Q_0t^4e^{-\frac{t}{K}}}{24K^4}$$

n = 6

$$I \left( 1 - e^{-\frac{t}{K}} - \frac{te^{-\frac{t}{K}}}{K} - \frac{t^2e^{-\frac{t}{K}}}{2K^2} - \frac{t^3e^{-\frac{t}{K}}}{6K^3} - \frac{t^4e^{-\frac{t}{K}}}{24K^4} - \frac{t^5e^{-\frac{t}{K}}}{120K^5} \right) + \frac{Q_0t^5e^{-\frac{t}{K}}}{120K^5}$$

n = 7

$$I \left( 1 - e^{-\frac{t}{K}} - \frac{te^{-\frac{t}{K}}}{K} - \frac{t^2e^{-\frac{t}{K}}}{2K^2} - \frac{t^3e^{-\frac{t}{K}}}{6K^3} - \frac{t^4e^{-\frac{t}{K}}}{24K^4} - \frac{t^5e^{-\frac{t}{K}}}{120K^5} - \frac{t^6e^{-\frac{t}{K}}}{720K^6} \right) + \frac{Q_0t^6e^{-\frac{t}{K}}}{720K^6}$$

From these solutions, we can easily identify the general pattern that can be expressed as:

$$Q^{(n)}(t) = I - Ie^{-t/K} \sum_{i=0}^{n-1} \frac{(t/K)^i}{i!} + Q_0e^{-t/K} \sum_{i=0}^{n-1} \frac{(t/K)^i}{i!}$$

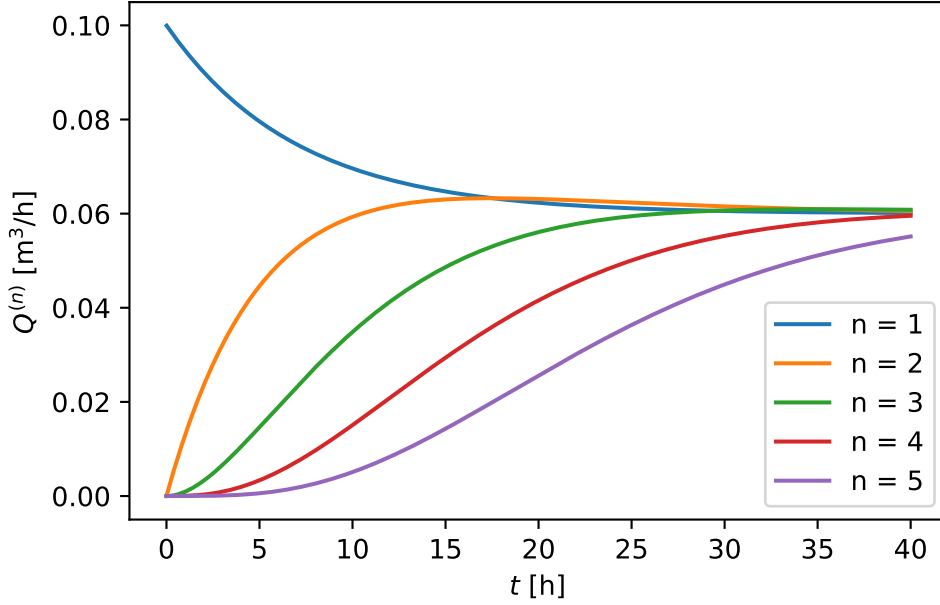
which is identical to the formulation in the theory notes. We can now also easily plot results for different  $n$  values

```

I_num = 0.06 # m^3/h
n_num_list = [1, 2, 3, 4, 5]
fig, ax = plt.subplots()
for n_num in n_num_list:
    Q_n_solved_specific_ = sp.simplify(Q_n_solved.subs(n, n_num))
    Q_n_func_ = sp.lambdify(
        t, Q_n_solved_specific_.subs({I: I_num, K: K_num, Q0: Q_0_num}),
        "numpy"
    )
    Q_n_ = Q_n_func_(t_array)
    ax.plot(t_array, Q_n_, label=f"n = {n_num}")

```

```
ax.set_xlabel(r"$t$ [h]")
ax.set_ylabel(r"$Q^{(n)}$ [m$^3$/h]")
ax.legend()
```



The larger  $n$ , the more delayed the response of the outflow is.

#### 1.1.4 Nash-cascade

A Nash-cascade is a special case of the  $n$ -th order linear reservoir, where the input  $I$  to the first reservoir is a [Dirac delta function](#)  $\delta(t)$ :

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

with

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

It has the very useful property that  $\mathcal{L}\{\delta(t)\} = 1$ <sup>1</sup>. Additionally assuming that the first reservoir is initially empty, i.e.  $Q^{(1)}(0) = 0$ , the continuity equation for the first reservoir of Equation 2 sets  $I = \delta(t)$  and  $Q(t=0) = 0$ . The ODE can be laplace transformed:

<sup>1</sup>Note that  $t$  needs to be defined between  $-\infty$  and  $\infty$  for this to be true

```

t = sp.symbols("t") # Generalise t for correct dirac delta transform!
I_dirac = sp.DiracDelta(t)
ode_linear_nash_1 = K * sp.diff(Q(t), t) - (I_dirac - Q(t))
ode_linear_nash_1_laplace = sp.laplace_transform(ode_linear_nash_1, t, s,
    ↪ noconds=True)
ode_linear_nash_1_laplace =
    ↪ sp.laplace_correspondence(ode_linear_nash_1_laplace, {Q: q})
display(ode_linear_nash_1_laplace)

```

$$K(sq(s) - Q(0)) + q(s) - 1$$

where we can clearly see that the Laplace transform of the Dirac delta function is 1. Inserting  $Q(0) = 0$  gives and solving for  $q(s)$  gives:

```

ode_linear_nash_1_laplace_ic = sp.laplace_initial_conds(
    ode_linear_nash_1_laplace, t, {Q: [0]}
)
q_1_nash = sp.solve(ode_linear_nash_1_laplace_ic, q(s))[0]
display(q_1_nash)

```

$$\frac{1}{Ks + 1}$$

So in the case of a dirac delta input, also the first reservoir has a transfer function of the same form as the other reservoirs (see Equation 8).

$$H^{(1)}(s) = \frac{q_1(s)}{I(s)} = \frac{q_1(s)}{1} = \frac{1}{1 + Ks}$$

Therefore, Equation 9 can be simplified to:

$$q^{(n)}(s) = \prod_{i=1}^n H^{(i)}(s) = \frac{1}{(1 + Ks)^n}$$

Transforming back to the time domain gives:

```

q_n_s_nash = 1 / ((1 + K * s) ** n)
Q_n_solved_nash = sp.simplify(sp.inverse_laplace_transform(q_n_s_nash, s, t))
Q_n_solved_nash

```

$$\frac{K^{-n}t^{n-1}e^{-\frac{t}{K}}\theta(t)}{(n-1)!}$$

In the above equation,  $\theta(t)$  is the Heaviside function (cf. supra for definition). As we are only interested in  $t \geq 0$  (i.e. after the input is applied) we can simplify the solution to:

```
Q_n_solved_nash_simplified = Q_n_solved_nash.subs(
    sp.Heaviside(t), sp.Heaviside(t).rewrite(sp.Piecewise)
)
Q_n_solved_nash_simplified = sp.refine(Q_n_solved_nash_simplified,
    ↪ sp.Q.positive(t))
Q_n_solved_nash_simplified
```

$$\frac{K^{-n}t^{n-1}e^{-\frac{t}{K}}}{(n-1)!}$$

Again note the equivalence to the formulation in the theory notes. Interestingly, the solution here assumes an empty first reservoir and a Dirac delta input, while the theory notes assume no input and a first and  $Q^{(1)}(t=0) = 1$ . The two assumptions are equivalent, as the Dirac delta input can be interpreted as a unit input to an empty reservoir.

Because the Nash-cascade is the response to an instantaneous input, it can be interpreted as an Instantaneous Unit Hydrograph (IUH)! Also note that the assumption that  $n$  is an integer can be relaxed to  $n$  being a positive real number. In this case, applying the inverse Laplace transform yields:

```
n = sp.symbols("n", positive=True, real=True)
q_n_s_nash = 1 / ((1 + K * s) ** n)
Q_n_solved_nash = sp.simplify(sp.inverse_laplace_transform(q_n_s_nash, s, t))
Q_n_solved_nash
```

$$\frac{K^{-n}t^{n-1}e^{-\frac{t}{K}}\theta(t)}{\Gamma(n)}$$

This allows  $n$  to be calibrated as a real number to match the geomorphological characteristics of a catchment. For completeness, we visualise the Nash-cascade for different  $n$  values, including non-integer values:

```
n_num_list = [1, 1.5, 2, 2.5, 3, 5]
fig, ax = plt.subplots()
for n_num in n_num_list:
    Q_n_solved_specific_ = sp.simplify(Q_n_solved_nash.subs(n, n_num))
    Q_n_func_ = sp.lambdify(t, Q_n_solved_specific_.subs({K: K_num}),
    ↪ "numpy")
```

```

    Q_n_ = Q_n_func_(t_array)
    ax.plot(t_array, Q_n_, label=rf"n = {n_num}")
ax.set_xlabel(r"$t$ [h]")
ax.set_ylabel(r"$Q$ [m$^3$/h]")
ax.legend()

```

